

ID No: _____ Name: _____ Sec No.: _____

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI

II SEMESTER 2025-26

CS/ECE/EEE/INSTR F241 Microprocessor Programming & Interfacing

11-02-2026 (Wednesday)

QUIZ-1 (closed book)

A

Max time: 20 Min

Max Marks: 08

Q1. Read the snippet of the program. Data Memory is initialized from offset 0120h and DS=CS=ES=SS=0863, , DI = 020BH, AX=BX=CX=DX=SI=DI = 0000H, Assume all flags are cleared initially (i.e '0'). [1M*8=8M]

Note: Read the question and then answer- do not spend time understanding the whole program

.MODEL TINY

.DATA

HEAD1 DB 1AH, 2BH

ALIGN 2

HEAD2 DB 3CH, 4DH, 5EH

STR1 DB 61H, 62H, 63H

STR2 DB 3 DUP(0FFH)

ARR1 DW 7FFFH

ALIGN 2

ARR2 DW 1 DUP(?)

MID1 DB 12H, 34H, 56H

MID2 DB 78H, 9AH, 0BCH

STR3 DB 3 DUP(?)

.CODE

.STARTUP

MOV SI, OFFSET ARR1

MOV BX, [SI]

INC WORD PTR [SI]

ADC BX, 0001H

MOV ARR2, BX

MOV SI, OFFSET STR1+2

LEA DI, STR2

MOV CX, 3

STD

REP MOVSB

.EXIT

END

Question	Answer (in hex)
(a) What is the value stored at memory location DS:130 after the program is loaded (before execution)?	12
(b) What is the 16-bit value stored at DS:(OFFSET ARR1) immediately after the instruction <code>INC WORD PTR [SI]</code> executes?	8000
(c) What are the final values of SI & DI after the execution of the program?	SI: 124 DI:125
(d) What is the value (word) stored in ARR2 after execution?	8000
(e) What is the value of the overflow Flag after ADC?	1
(f) Mention the value of 3 bytes of memory starting from STR1 (3 bytes) after the execution of the instructions? Mention the three bytes in order starting from the one in lowest address.	61,61,62
(g)What are the final contents of STR2 (3 bytes) after the execution of the instructions? Mention the three bytes in order starting from the one in lowest address.	63,FF, FF
(h) What is the offset of ARR2?	12E

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B

Max time: 20 Min

Max Marks: 08

Q1. Read the snippet of the program. Data Memory is initialized from offset 0120h and DS=CS=ES=SS=0863, , DI = 020BH, AX=BX=CX=DX=SI=DI = 0000H, Assume all flags are cleared initially (i.e '0'). [1M*8=8M]

Note: Read the question and then answer- do not spend time understanding the whole program

.MODEL TINY

.DATA

HEAD1 DB 10H, 20H

ALIGN 2

HEAD2 DB 30H, 40H, 50H

STR1 DB 41H, 42H, 43H

STR2 DB 3 DUP(01)

ARR1 DW 0FFFFH

ALIGN 2

ARR2 DW 1 DUP(?)

MID1 DB 60H, 70H, 80H ; 3 bytes

MID2 DB 90H, 0A0H, 0B0H ; 3 bytes

STR3 DB 3 DUP(?)

.CODE

.STARTUP

MOV SI, OFFSET ARR1

MOV BX, [SI]

INC WORD PTR [SI]

ADC BX, 0001H

MOV ARR2, BX

MOV SI, OFFSET STR1+2

LEA DI, STR2

MOV CX, 3

STD

REP MOVSB

.EXIT

END

Question	Answer (in hex)
(a) What is the value stored at memory location DS:130 after the program is loaded (before execution)?	60
(b) What is the 16-bit value stored at DS:(OFFSET ARR1) immediately after the instruction <code>INC WORD PTR [SI]</code> executes?	1000
(c) What is the offset of ARR2?	12E
(d) What is the value (word) stored in ARR2 after execution?	1000
(e) What is the value of the overflow Flag after ADC?	0
(f) Mention the value of 3 bytes of memory starting from STR1 (3 bytes) after the execution of the instructions? Mention the three bytes in order starting from the one in lowest address.	41,41,42
(g) What are the final contents of STR2 (3 bytes) after the execution of the instructions? Mention the three bytes in order starting from the one in lowest address.	43,01,01
(h) What are the final values of SI and DI?	SI: 12A DI: 125